

1. Administrative & Study Identification

Full Study Title: Substudy 03C: A Study of Combination Therapies in Participants With Renal Cell Carcinoma With Recurrent Disease During or After Anti-PD-(L)1 Therapy (MK-3475-03C/KEYMAKER-U03)
Short Title/Acronym: KEYMAKER-U03 Substudy 03C **Protocol Number:** MK-3475-03C **NCT Number:** NCT07049926 **Trial Status:** Recruiting
Sponsor Name: Merck & Co., Inc. (MSD) **Investigational Product:** Belzutifan (Welireg) plus Zanzalintinib; Belzutifan (Welireg) monotherapy **Phase of Development:** Phase 1b/2 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To establish the recommended phase 2 dose (RP2D) of belzutifan plus zanzalintinib during the safety lead-in phase, and to evaluate the Objective Response Rate (ORR) per RECIST v1.1 by Blinded Independent Central Review (BICR) in the efficacy phase. **Secondary Objectives:** To evaluate clinical benefit rate, duration of response (DOR), progression-free survival (PFS) per RECIST v1.1 by BICR, and overall survival (OS).

2.2 Background & Rationale Adjuvant pembrolizumab is a standard of care for patients with renal cell carcinoma (RCC) at intermediate-high or high risk of recurrence post-surgery. Novel treatment approaches for recurrence after adjuvant anti-PD-(L)1 therapy are urgently needed. This study is designed to evaluate the safety and efficacy of the HIF-2α inhibitor belzutifan combined with the tyrosine kinase inhibitor zanzalintinib (or belzutifan monotherapy) for recurrent clear cell RCC (ccRCC) during or after anti-PD-(L)1 adjuvant therapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Belzutifan monotherapy arm) **Blinding:** Open-label **Randomization:** 2:1 ratio (belzutifan + zanzalintinib vs. belzutifan monotherapy) during the efficacy phase

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 140 participants (10 in safety lead-in, 120 in efficacy phase, plus additional rolling allocations) **Number of Sites:** Global Multicenter **Estimated Study Start:** [Insert Date] **Estimated Primary Completion:** [Insert Date]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with Karnofsky performance status of $\geq 70\%$. 2. Histologically confirmed unresectable locally advanced or metastatic clear cell renal cell carcinoma (ccRCC). 3. Measurable disease per RECIST v1.1 as assessed by investigator and verified by BICR. 4. Disease recurrence during adjuvant anti-PD-(L)1 therapy or ≤ 24 months following the last dose of adjuvant anti-PD-(L)1 therapy. 5. Submits an archival tumor tissue sample or newly obtained core/excisional biopsy of a tumor lesion not previously irradiated.

4.2 Exclusion Criteria 1. Has a known additional malignancy that is progressing or required active treatment within the past 3 years. 2. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 3. Radiographic evidence of encasement or invasion of a major blood vessel, or intratumoral cavitation. 4. Moderate to severe hepatic impairment or gastrointestinal (GI) disorders associated with high risk of perforation, fistula, or malabsorption. 5. Currently receiving anticoagulants or platelet inhibitors that cannot be discontinued for the study duration.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * *Safety Lead-in Phase:* Belzutifan 120 mg orally once daily plus zanzalintinib 60 mg or 100 mg orally once daily. * *Efficacy Phase:* Belzutifan 120 mg orally once daily plus zanzalintinib at the RP2D, OR belzutifan 120 mg orally once daily monotherapy. **Route of Administration:** Oral **Duration of Treatment:** Continues until documented disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Bone resorptive therapy (must be initiated at least 2 weeks before allocation/randomization); antihypertensive medications. **Prohibited:** Anticoagulants or platelet inhibitors that cannot be discontinued; prior systemic treatment for advanced/metastatic disease except for adjuvant anti-PD-(L)1 therapy (and max 1 prior VEGF TKI in the safety lead-in phase).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy Collection, Baseline Scans
Baseline	Day 0	Allocation/Randomization, First Dose of Study Intervention
Interim	Every [X] Weeks	Safety Labs, Efficacy Assessment (RECIST v1.1 scans), AE Monitoring
End of		

Study | Variable | Disease Progression or Unacceptable Toxicity,
Exit Interview, Survival Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Establishment of the Recommended Phase 2 Dose (RP2D) (Safety Phase) and Objective Response Rate (ORR). **Measurement Method:** Assessed per RECIST v1.1 by Blinded Independent Central Review (BICR).

7.2 Secondary & Exploratory Endpoints * Clinical Benefit Rate * Duration of Response (DOR) per RECIST v1.1 by BICR * Progression-Free Survival (PFS) per RECIST v1.1 by BICR * Overall Survival (OS)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of adverse events and dose-limiting toxicities (DLTs) to establish the RP2D. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal sponsor review committee to evaluate the safety lead-in phase data prior to opening the efficacy phase.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy outcomes; Safety Set for all participants receiving at least one dose of study treatment. **Sample Size Justification:** Approximately 140 participants (\$N=10\$ safety lead-in; \$N=120\$ efficacy phase randomized 2:1). No hypothesis testing and no formal comparisons are planned. **Handling of Missing Data:** Standard rules apply for time-to-event endpoints (censoring at last known alive/progression-free date).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence. **Audit Trail:** Electronic Case Report Form (eCRF) locking and standard query resolution workflows. Blinded Independent Central Review (BICR) will be utilized to audit tumor response data.

11. Study Identifiers & Governance

Primary ID: MK-3475-03C **Registry Number:** NCT07049926 **Sponsor/Lead Organization:** Merck & Co., Inc. **Study Title:** KEYMAKER-U03 Substudy 03C **Official Regulatory Title:** Substudy 03C: A Study of Combination Therapies in Participants With Renal Cell Carcinoma With Recurrent Disease During or After Anti-PD-(L)1 Therapy (MK-3475-03C/KEYMAKER-U03)

12. Clinical Framework (Oncology Specific)

Primary Condition: Clear Cell Renal Cell Carcinoma (ccRCC) **Diagnosis Threshold:** Histologically confirmed unresectable locally advanced/metastatic ccRCC **Medical Methodology:** Targeted combination therapy (HIF-2α inhibitor + TKI) vs. targeted monotherapy **Optimization Target:** Overcoming resistance in recurrent disease after anti-PD-(L)1 adjuvant therapy

13. Study Procedures & Methodology

Management Pathway: Standard oncology treatment pathway with rigid tumor assessment intervals. **Education Protocol (HCP):** RECIST v1.1 training, AE/DLT management guidelines. **Education Protocol (Patient):** Oral medication adherence, side-effect self-reporting, blood pressure monitoring at home. **Quality Audit Mechanism:** Independent imaging review (BICR) to prevent investigator bias in ORR/PFS measurements.

14. Observation & Follow-Up Schedule

Total Duration: Dependent on PFS/OS events (patients followed until death or study closure) **Onsite Visit Cadence:** Every cycle (typically 3-4 weeks) for drug dispensation and safety checks. **Remote Monitoring Frequency:** Routine telemedicine checks for AE monitoring as applicable. **Checklist Review Interval:** Tumor imaging (CT/MRI) conducted at regular intervals per protocol RECIST v1.1 requirements.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				